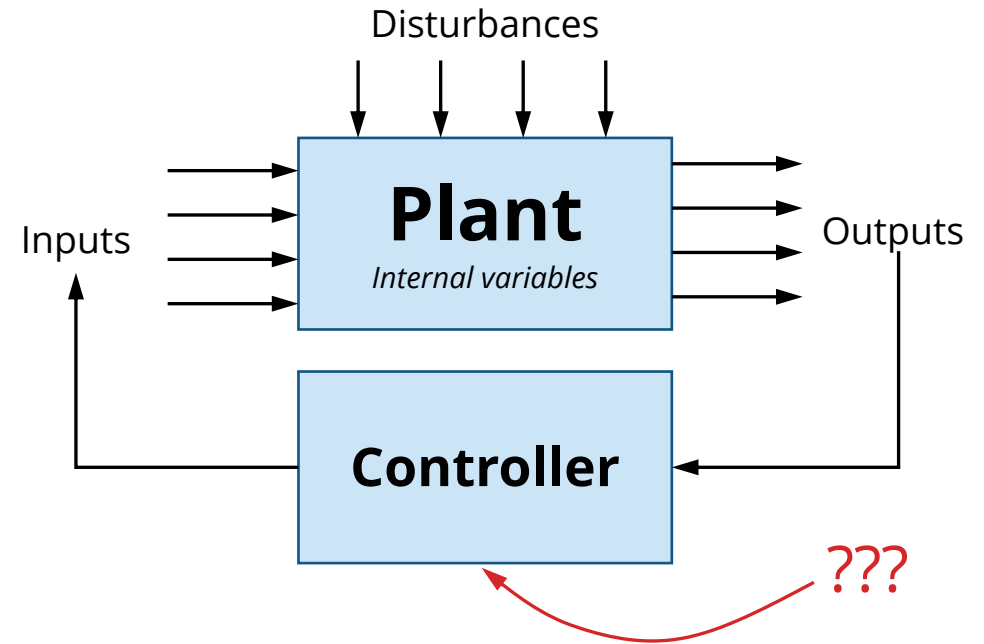




AA 513 Multivariable Control

Lecture 02 // Week 2

Instructor: Prof. Karen Leung

Announcements

- More exercises are due in week 3 (exercises from weeks 1 & 2)
- Clarification on “tutorial”
 - A slight variation to the typical “course project”
 - Recording of you presenting slides on the project details
 - Instead of a technical report, you provide a notebook with text + code + figures.
- TA OH set up Wed 6PM
- Canvas course email corrected
- Extra credit for Ed Discussion participation

Last time

- Introduction to state space representation
 - States (set of variables sufficient to describe the system)
 - Control input (we decide the value of it)
 - Disturbance input (can be used to capture uncertainty in the system)
 - Dynamics (how the states evolve over time)
 - Continuous time vs discrete time
 - Nonlinear vs control affine vs linear
 - Linearizing dynamics
- Coding
 - Basics of defining dynamics and simulating trajectories
 - Euler vs RK4
 - JAX (`vmap`, `jit`, `grad`, `jacobian`, `hessian`)

This week

- Finish up linearization
- Introduction to optimization
- Exercise 2 & 3

Review: Gradients, Jacobian, Hessians

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Introduction to optimization

What is optimization?

- **In words:** The process of finding the *best* decision or design according to a quantitative criterion, subject to constraints that describe what is possible.

- **In math:** $\min_x f(x)$ Measures performance

$$\text{s.t. } g(x) \leq 0$$

Defines feasibility

$$h(x) = 0$$

Is x the state?

- **Why?** Provides a systematic way to make decisions. Instead of producing a function or rule to make decisions, we can instead articulate what we want and let an algorithm decide.
- **Connection to controls:** Find best control input to accomplish desired task subject to dynamics and constraints

Applications of optimization

Energy and infrastructure (*Civil/mechanical engineering*)

- *Power grid operation*: optimize generator outputs to meet demand at minimal cost (economic dispatch).
- *Building HVAC systems*: minimize energy while maintaining comfort constraints.

Finance & Business (*Operations research, economics*)

- *Portfolio optimization*: allocate assets to maximize return for a given risk.
- *Supply chain management*: minimize cost of production, transport, and inventory.

Science & Health (*Biology, health sciences, medicine*)

- *Drug dosage optimization*: find schedules that maximize efficacy and minimize side effects.
- *Epidemic control*: determine optimal vaccination or intervention strategies.

Machine Learning

- *Supervised learning*: Given a dataset, find parameters of a model that minimizes prediction accuracy

Aerospace applications

Discuss: Objectives and constraints

- Aircraft design and performance
 - Aerodynamics
 - Structures
 - Propulsion
- Spacecraft mission planning
 - Orbit transfer
 - Attitude control
 - Constellation design
 - Mission scheduling
- Operations and logistics
 - Air traffic management
 - Fleet and maintenance scheduling

Aerospace *control* applications

- Flight trajectory and guidance (high-level planning)
 - Trajectory/path planning
 - Landing locations
 - Formation flight
- Control and estimation (navigation)
 - Tracking a trajectory
 - Estimating state

Worst-case analysis

- We can use optimization to find the *worst-case* (pessimization)
- Why?
 - Assess robustness. Still tolerable under worst-case?
 - Falsification / find counter example
 - Collect adversarial data to train model

How to solve optimization problems?

$$\min_x f(x)$$

Measures performance

$$\text{s.t. } g(x) \leq 0$$

Defines feasibility

$$h(x) = 0$$

- From your calculus class: *Find stationary points and then ...*

Example: unconstrained

- $f(x) = (x - 3)^2$ (no constraints)

Example: constrained, Lagrangian multipliers

- $f(x) = x^2, x \geq 1$

Necessary and sufficient conditions

- **Necessary** “must have” but by itself is not enough

$$\nabla f(x^*) = 0$$

(More generally, Karush-Kahn-Tucker conditions)

- **Sufficient**, enough to show optimality
 - $\nabla f(x^*) = 0$
 - Convexity

Gradient descent

- What if finding stationary points is hard?
 - $f'(x) = 0$ is difficult to solve analytically (see exercise 3)

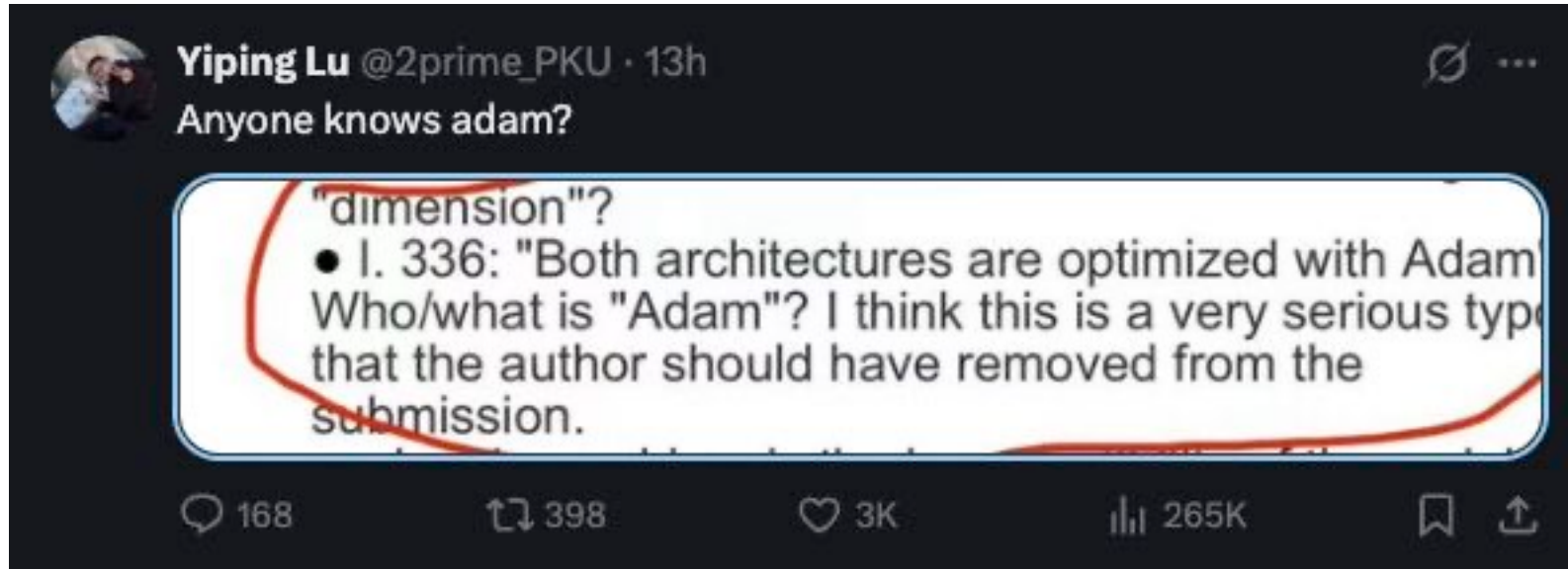
Gradient descent: Practical considerations

- Step size
- Initial conditions
- Differentiability of the function

More advanced gradient descent algorithms

- Methods to adjust the step size as a function of function landscape
 - **Momentum**: adds inertia, uses previous gradients
 - **Nesterov Accelerated Gradient (NAG)**: Incorporates lookahead
 - **AdaGrad**: uses gradient history
 - **RMSProp**: Improvement on AdaGrad
 - **Adam**: Momentum + RMSProp
 - **Second-order methods**
 - **Line search**: Another optimization problem
- Trade-off between convergence, complexity, computation cost

Side note



- NeurIPS: One of the largest AI/ML research conference
- Some concerns regarding review quality

A whole field on (analytic) optimization theory and tools

- Linear programming (LP)
- Quadratic program (QP)
- Convex optimization (CVX)
- Quadratic constrained quadratic program (QCQP)
- Mixed integer linear program (MILP)
- Nonlinear optimization (NL)
- Lagrangian optimization
- Duality
- Quasi-Newton methods
- ...

How to solve optimization problems?

$\min_x f(x)$ Measures performance

s.t. $g(x) \leq 0$
 $h(x) = 0$ Defines feasibility

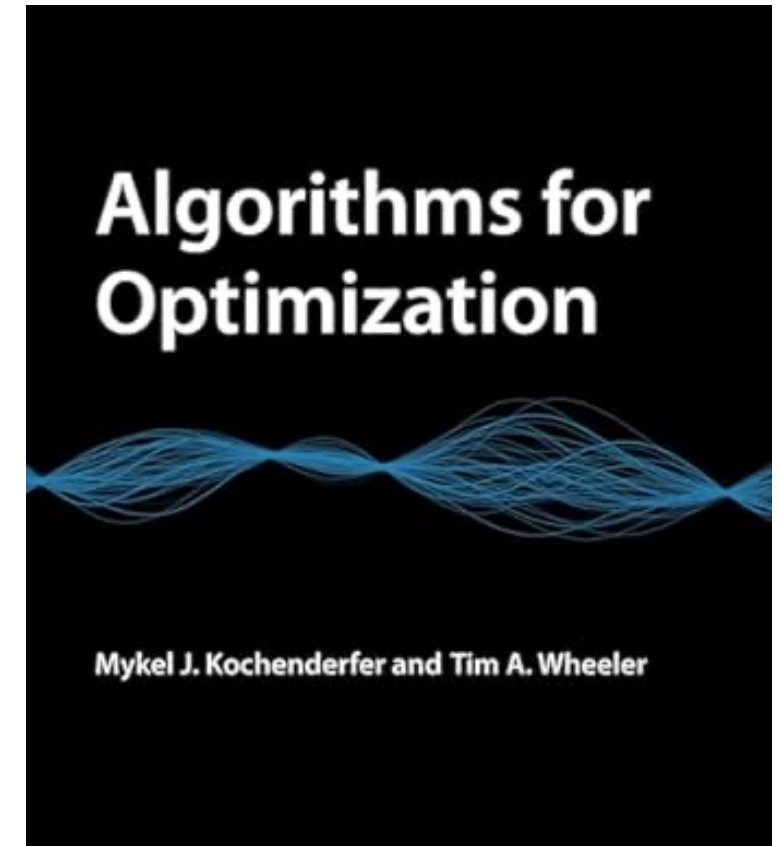
- Depends on how f, g, h are evaluated
 - Differentiable?
 - Analytic?
 - Cheap or expensive?
 - Known or black box?

Anti-examples of gradient descent

- Minimize drag on aircraft body
 - Objective = drag
 - Need CFD (black-box) or wind tunnel testing (expensive)
- Choose best wing geometry that minimize drag
 - Straight, swept, delta, elliptical, variable geometry (discrete)
 - Number of wings (discrete)
 - Span, chord, camber (continuous)

So many other optimization algorithms

- Stochastic method
 - Cross-entropy method
 - Simulated annealing
- Population methods
 - Genetic algorithm
 - Cuckoo search
 - Particle swarm optimization
- Surrogate models
- ...



Algorithms for Optimization <https://mykel.kochenderfer.com/textbooks/>

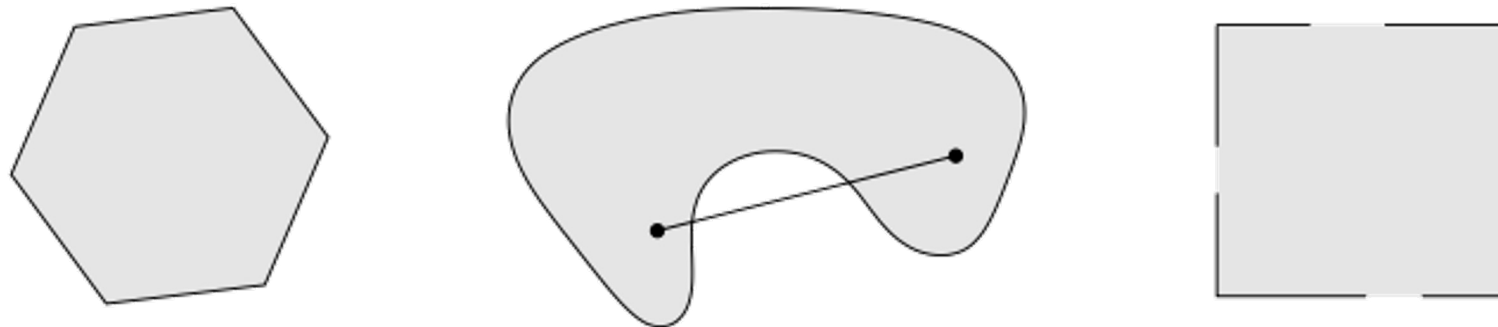
Convex optimization

- A class of optimization problem
 - There are established tools to solve sub-classes of convex optimization reasonably well.
 - (Or at least have hope of finding a reasonably good solution)
- Formulating an optimization problem as convex is the bulk of the challenge.

https://web.stanford.edu/~boyd/cvxbook/bv_cvxslides.pdf

Convex sets

- **Line segment:** line connecting two points
- **Convex set:** contains line segment between any two points in the set.



From EE 364A lecture notes

Hyperplane and half spaces

- **Hyperplane:** set of the form $\{x \mid a^T x = b\}, a \neq 0$
- **Halfspace:** set of the form $\{x \mid a^T x \leq b\}, a \neq 0$

Question

- What does $\{x \mid Ax \leq b\}$, $A \neq 0, A \in R^{m \times n}$ describe?

Convex functions

- $f: R^n \rightarrow R$ is convex if $\text{dom } f$ is a convex set, and for all $x, y \in \text{dom } f, 0 \leq \theta \leq 1 \dots$

Examples of convex functions

- On R
 - Affine
 - Exponential
 - Quadratic
- On R^n
 - Affine
 - Norms
 - Sum of squares / quadratic

First-order conditions

- Assuming gradient exists
- First order Taylor approximation (tangent) is a global under-estimator
- Differentiable f with convex domain is convex if and only if

Second order conditions

- Assuming twice differentiable
- Twice differentiable f with convex domain is convex if and only if $\nabla^2 f(x) \succcurlyeq 0$
 - $\succcurlyeq$ not $\geq$
 - Positive (semi-)definite
- Relationship with eigen values?

Convex optimization problem

$$\min_x f(x)$$

Convex objective

$$\text{s.t. } g_i(x) \leq 0, \quad i = 1, \dots, m$$

Convex inequality constraints

$$a_i^T x = b_i, \quad i = 1, \dots, p$$

Affine equality constraints

Quadratic program

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^T P x + q^T x + r \\ \text{s.t.} \quad & Gx \leq h \\ & Ax = b \end{aligned}$$

Recurring theme: Turn problems into a QP

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^T P x + q^T x + r \\ \text{s.t.} \quad & Gx \leq h \\ & Ax = b \end{aligned}$$

Exercises

- Gradient descent
- Unconstrained optimization
- Constrained optimization
- CVXPY – python package for solving convex optimization problems